

What Musical Features Drive Song Popularity? A Spotify Audio Feature Analysis Over Time

Albert Skalinski

MSc in Data Analytics, DkIT

Objectives

The main aim of this Dissertation is to explore whether Spotify audio features can be used to understand and predict song popularity. The project is guided by two research questions:

- How have the structural determinants of song popularity evolved over time in contemporary music?
- How accurately can song popularity be predicted using Spotify audio features?

Background

Music popularity is difficult to explain because it depends both on measurable track characteristics and external factors. Spotify provides audio features such as energy, danceability, loudness, acousticness, instrumentalness, and many more. They make it possible to analyse songs quantitatively.

Although, popularity is not determined by sound only. There is a number of factors that also might affect whether a track becomes popular or not: artist recognition, playlist placement, promotion, or social media trends.

Data

The Dissertation uses a publicly available Kaggle Spotify music dataset. It contains descriptive track information and Spotify audio features.

The descriptive variables include:

- track name
- artist name
- album name
- release date
- playlist genre/subgenre
- popularity score

The audio features include:

- danceability
- energy
- loudness
- acousticness
- instrumentalness
- valence
- speechiness
- liveness
- tempo

Methodology

This project follows the following pipeline:

- Data preparation** - two original .csv files were loaded, inspected, and joined into one dataset
- Data cleaning** - data types, missing values, duplicates and the general quality of data were assessed. Incomplete or problematic dates were handled
- Feature engineering** - new temporal features were created, including releaseYear and releaseMonth. A popularityGroup variable was also created to divide the dataset into low, medium, and high popularity tracks
- EDA** - the following were explored during this stage: distributions, relationships, correlations, and differences between popularity groups
- Planned modelling** - the next stage will involve regression models and feature importance analysis across different time periods

Current Progress

The following work has been completed so far:

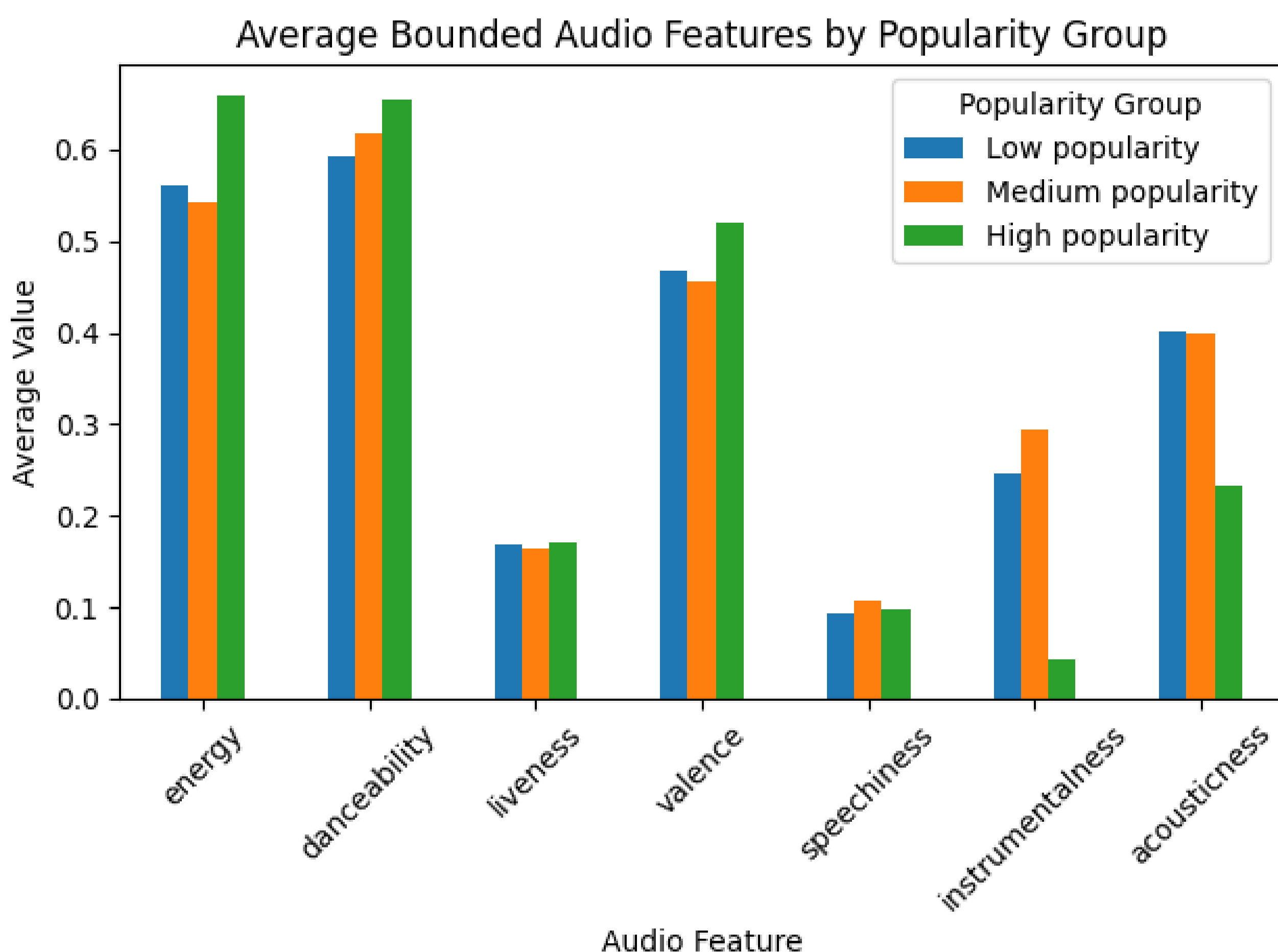
- Dataset files have been joined
- Missing/incomplete dates have been cleaned
- Release year and month features have been created
- Popularity groups have been created
- EDA has been completed
- Initial visualisations have been produced
- The artefact pipeline has been designed

Preliminary Results

The EDA produced several initial findings:

- The dataset is **strongly skewed towards recent music** - older songs are less represented. Older tracks that remain popular on Spotify may not represent older music as a whole
- Most individual **audio features do not show a strong relationship with popularity** - song popularity is most likely a multivariate case rather than something that could be explained by one feature alone
- The strongest **positive relationship** is between **loudness and energy** - energetic tracks are often louder and more intense
- Popularity showed negative relationships with instrumentalness and acousticness** - highly popular tracks are generally less instrumental and acoustic

Overall, the results suggest that Spotify audio features may help explain popularity, but they are **unlikely to fully explain it without external factors**, such as artist popularity, or promotion.



Planned Future Work

The next stage of the Dissertation will focus on predictive modelling and temporal interpretation.

The dataset will be divided into time periods based on the release year. This will help in comparing whether the importance of the features changes over time.

A baseline regression model will then be created. After that, a more advanced model (e.g., XGBoost) will be tested to see whether it improves the performance.

Different performance metrics will be evaluated, such as MAE, RMSE, or R^2 . Finally, feature importance will be compared across different time periods. This will help to answer the question of whether structural determinants of song popularity have changed over time, or not.

Expected Contribution

This project contributes to the field by applying a full analytical pipeline to cultural and entertainment data. It demonstrates how music can be explored using quantitative features, visualisations, and predictive modelling. The project is expected to show that Spotify audio features can partially predict popularity, but that their predictive power is limited, as popularity also depends on external factors. It is also expected that the importance of musical features will change across time periods.

References

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- The above references have been generated with the aid of this tool: <https://www.mybib.com/tools/harvard-referencing-generator>

Dataset used: <https://www.kaggle.com/datasets/solomonameh/spotify-music-dataset/data>